# *C* and von Neumann Algebras*

## Directed Reading Program in Mathematics


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## *Preliminary Definitions*

- A Banach algebra is a complete normed vector space equipped with binary multiplication.
- An involution on an algebra $A$ is a map $a \mapsto a^*$ from $A$ to itself such that:
  - $(a^*)^* = a$
  - $(ab)^* = b^* a^*$
  - $(a + b\lambda)^* = a^* + \overline{\lambda} b^*$ for all $a, b \in A, \lambda \in \mathbb{C}$
- A C* algebra is a Banach algebra with an involution such that $||a^* a|| = ||a||^2$.

Let $A$ be a commutative Banach algebra and let $\Delta(A)$ be the set of nonzero complex algebra homomorphisms $h : A \to \mathbb{C}$. We call this the *spectrum* of $A$.

## *The Gelfand Representation*

Given an abelian Banach Algebra, $A$, for any $a \in A$ define the Gelfand map $a \mapsto \hat{a}$ where $\hat{a} : \Delta(A) \to \mathbb{C}$ is defined by $\hat{a}(h) = h(a)$

- By definition of $\Delta$ this is an algebra homomorphism.
- The set $\Delta$ is compact and Hausdorff.
- The image of the Gelfand map is a subset of $C_0(\Delta)$.

Let $A$ be an abelian Banach Algebra. The map $\Phi : A \to C_0(\Delta)$ defined by $\Phi(a) = \hat{a}$ is an norm-decreasing homomorphism.

$$||\hat{a}|| = \rho(a) = \lim_{n \to \infty} ||a^n||^{\frac{1}{n}} \leq ||a||$$

In the case that $A$ is an abelian C* algebra, using the C* norm property, it can be shown that $\Phi$ is an isometry.

Theorem: Let $A$ be an abelian C* algebra. Then $\Phi$ is an isometric *-isomorphism from $A$ onto $C_0(\Delta)$.

## *Examples*

Example 1: $C(X)$ for $X$ compact and Hausdorff.

Spectrum

$X$

Gelfand Map

$\Phi(f)(x) = f(x)$

$\Phi(C(X)) = C(X)$

Example 2: $C^*(1, a)$ for $a \in A$ normal.

Spectrum

$\sigma(a) = \{\lambda \in \mathbb{C} : a - \lambda \cdot 1 \text{ is not invertible}\}$

Gelfand Map

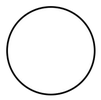
$\Phi(a)(\lambda) = \lambda$

$\Phi(C^*(1, a)) = C(\sigma(a))$

Example 3: The space of absolutely summable sequences indexed by the integers, $\ell^1(\mathbb{Z})$. This is a Banach algebra with pointwise addition and circular convolution.

Spectrum

$S^1$



Gelfand Map

$$\Phi(\{c_n\})(z) = \sum_{n=-\infty}^{\infty} c_n z^n$$

$\Phi(\ell^1(\mathbb{Z})) \subset C(S^1)$

Example 4: The space of Lebesgue integrable functions. This is a Banach algebra with pointwise addition and convolution.

Spectrum

$\mathbb{R}$

$\longleftrightarrow$

Gelfand Map

$$\Phi(f)(y) = \int_{\mathbb{R}} f(x) e^{-ixy} dx$$

$\Phi(L^1(\mathbb{R})) \subset C_0(\mathbb{R})$

Non-Example: Bounded operators on a Hilbert Space, $\mathcal{B}(\mathcal{H})$. This is a C* algebra, but it is not a commutative C* algebra so the Gelfand representation does not apply.

## *Theorem (Abelian von Neumann Algebras)*

Definition: A von Neumann algebra is a C*-subalgebra of $\mathcal{B}(\mathcal{H})$ that is closed in the strong operator topology.

Theorem: Let $A$ be an abelian von Neumann algebra acting on a Hilbert space $\mathcal{H}$. Let $x$ be a vector such that $\overline{[Ax]} = \mathcal{H}$.

Then, there exists a second countable compact Hausdorff space $\Omega$, a positive regular Borel complex measure $\mu$, and a unitary $u : \mathcal{H} \to L^2(\Omega, \mu)$ such that $uAu^*$ is the von Neumann algebra of all multiplication operators $M_f$ acting on $L^2(\Omega, \mu)$.

## *Proof of Theorem*

Step 1: Gelfand to construct $\Omega$

Let $B$ be a C*-subalgebra of $A$ that is strongly dense.
By Gelfand, there is $\Phi : B \to C(\Omega)$ where $\Phi$ is a *-isomorphism.

Step 2: Duality to construct $\mu$

Define $\tau : C(\Omega) \to \mathbb{C}$ via

$\tau(f) = \langle \Phi^{-1}(f)x, x \rangle$

(Riesz-Kakutani) $\Longrightarrow$ There exists $\mu$ with

$$\tau(f) = \int_{\Omega} f d\mu$$

Step 3: Define the action of $B$ on the Hilbert space $L^2(\Omega, \mu)$

We consider the following map:

$\pi : B \to \mathcal{B}(L^2(\Omega, \mu)), \quad b \mapsto M_{\Phi(b)}$

This map takes an element $b$ in our C*-algebra and sends it to multiplication by its Gelfand representation. In other words, we have $\pi(b)(f) = \Phi(b) \cdot f$

Step 4: Construct a unitary operator from $H$ to $L^2(\Omega, \mu)$

First consider the map

$u : Bx \to L^2(\Omega, \mu), \quad b(x) \mapsto \Phi(b)$

$\int |\Phi(b)|^2 d\mu = \tau(\Phi(b)^2) = \langle \Phi^{-1}\Phi(b^* b)x, x \rangle = ||bx||^2$

Therefore, $u$ is well-defined, isometric, and linear. Since $B$ is dense in $A$, we have that $\overline{[Bx]} = \mathcal{H}$. Therefore we can extend $u$ to a unitary from $\mathcal{H}$ to $L^2(\Omega, \mu)$

Step 5: The adjoint map and invoke density.

It can be shown that the adjoint map $\mathrm{Ad}_u : A \to \mathcal{B}(L^2(\Omega, \mu)), \quad a \mapsto uau^*$ is a *-isomorphism and is equal to $\pi$ on $B$. Therefore, from the Gelfand map,

$$\mathrm{Ad}_u(B) = \pi(B) = C(\Omega)$$

Finally, putting it all together,

$$\mathrm{Ad}_u(A) = \mathrm{Ad}_u(\overline{B}) = \overline{\mathrm{Ad}_u(B)} = \overline{C(\Omega)} = L^\infty(\Omega, \mu)$$

$B$ is strongly dense $\quad$ $\mathrm{Ad}_u$ is a homeomorphism $\quad$ By definition of $\pi$

## Final Remarks

- Certain topological properties can be formulated as algebraic properties on a C*-algebra $A$, without reference to any underlying space
  - This gives rise to "noncommutative topology"
- Similarly, properties of abelian von Neumann algebras correspond to measure theoretic properties
  - Noncommutative probability is called "free probability" and it has been a very rich and active area of research over the last couple decades